

Influence of mixture composition on the subcontinuum structure of reactive $\text{H}_2/\text{O}_2/\text{N}_2$ shock waves

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1) Novelty Statement

We quantify how mixture composition controls the subcontinuum structure of reactive shock waves in gaseous $\text{H}_2/\text{O}_2/\text{N}_2$ mixtures using reactive molecular dynamics. Beyond prior binary H_2/O_2 evidence [1], we quantify the composition-dependence of the separation of shock fronts of different species and chemical reactions in the shock region for nitrogen dilution which models hydrogen combustion in air, for two equivalence ratios. We show that the degree of shock separation follows molecular mass and are largest without dilution, then decreases as N_2 raises the upstream density. These results indicate that nitrogen promotes reaction intensification at the shock front, supporting faster local combustion in the shock front region in the detonation regime.

2) Significance Statement

High-pressure combustion of hydrogen fuels is gaining attention due to its high efficiency and low carbon footprint. Reactive shocks of $\text{H}_2/\text{O}_2/\text{N}_2$ mixtures govern ignition, detonation onset, and supersonic combustion in air-breathing systems for broad applications, yet microscopic reaction dynamics near the shock region remains challenging due to fast nonequilibrium thermochemistry and small shock thickness. Our study delivers first principle insights into the microscopic H_2/O_2 reactive shock physics in air and its dependence on the fuel equivalence ratio. These outcomes can be of critical use for the design of broad hydrogen-based combustion systems as well as for the validation of mesoscopic and continuum models.

3) Author Contribution

- TMO: Simulation, Formal analysis, Visualization, Writing, Investigation.
- KM: Formal analysis, Writing, Supervision, Investigation, Conceptualization, Funding acquisition.

4) Preferred Presentation Format

The authors request oral presentation format for the following reasons: a) the principal results are presented through layered visualisations that are most effectively communicated with accompanying oral explanation; b) the format enables concise clarification of methodology and interpretation of the coupled shock-chemistry dynamics, which is better achieved in an oral presentation than in a poster.

5) Colloquium/Colloquia Designation and Keywords

Colloquium choices : Detonations and high-speed combustion

Keywords: Hydrogen combustion; Molecular dynamics; Detonation; Nonequilibrium chemistry; Subcontinuum-scale dynamics

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Abstract

We use nonequilibrium reactive molecular dynamics (MD) simulations to analyze the subcontinuum structure and reaction dynamics of shock waves in gaseous $\text{H}_2/\text{O}_2/\text{N}_2$ mixtures. Our recent study of H_2/O_2 binary mixture [1] identified the separation of H_2 and O_2 shock fronts due to their large molecular mass disparity. To assess the effect of N_2 on the shock substructure, a piston-driven shock is simulated in a square duct for two mixtures at $\phi = 1.0$ and $\phi = 0.7$ with N_2/O_2 ratio of 4, where ϕ is the H_2/O_2 equivalence ratio. The separation of shock fronts are observed in both cases, ordered by the molecular mass of the species (H_2 leading, then N_2 , then O_2). The front offset is largest in the undiluted binary mixture and decreases when N_2 increases the upstream density. The reaction progress is quantified and evaluated through the shock front, and shows that reactions are enhanced with N_2 dilution. These findings show that composition plays a central role in governing how the shock interacts with chemical reactions, shaping both the structure and the early evolution of the reactive front at subcontinuum scales.

Keywords: Hydrogen combustion; Molecular dynamics; Reactive shock waves; Nonequilibrium dynamics; Subcontinuum structure

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1. Introduction

Hydrogen-air detonations are central to modern propulsion and power generation systems due to high efficiency of pressure gain combustion [2, 3] and low carbon footprint of hydrogen-based fuels. Applications include detonation engines and scramjet combustors [3–7]. Understanding shock-induced H_2/O_2 chemical reactions in the presence of nitrogen is essential for the design and improvement of these systems.

The presence of nitrogen in air can change the thermophysical properties of gaseous H_2/O_2 mixture as well as reaction dynamics [8]. The analysis of chemical reactions in the shock region however is challenging due to the fast nonequilibrium gas dynamics and the shock thickness as small as the molecular mean-free path below the continuum scale [9]. To address this challenge, we perform non-equilibrium reactive molecular dynamics (MD) simulations of near Mach 5 shock waves in gaseous $\text{H}_2/\text{O}_2/\text{N}_2$ mixtures, by extending our recent setup for H_2/O_2 mixtures [1].

The objective of this work is to quantify the influence of composition on the subcontinuum structure of reactive $\text{H}_2/\text{O}_2/\text{N}_2$ shock waves. Shock waves are generated by a moving piston in a square duct, with the piston speed chosen to target a nominal $M = 4$ in the initial state. We consider air-like nitrogen dilu-

tion and variations of the equivalence ratio ϕ to assess how dilution and mixture richness govern shock-front separation and post-shock reaction dynamics.

2. Methodology

The setup of reactive molecular dynamics (MD) simulations were conducted with LAMMPS [10] using the *ReaxFF* potential [11]. The detailed setups are available in prior work [1]. The binary H_2/O_2 case employs the CHO-2008 force field [12], while the ternary $\text{H}_2/\text{O}_2/\text{N}_2$ cases use the CHO-2016 force field [13] that also include nitrogen. *ReaxFF* explicitly models bond formation and dissociation, which enables tracking reactants, radicals, and products under the action of a shock generated by a moving piston.

The computational domain is rectangular in Cartesian coordinates x - y - z , initially filled with a gas mixture at $P_0 = 1$ atm and $T_0 = 500$ K, with zero mean velocity. The transverse cross-section is $0.025 \times 0.025 \mu\text{m}^2$, and the initial length is $L_z = 10 \mu\text{m}$. The x and y -boundaries are periodic. The piston is initially at $z = 0$, and the shock propagates in the $+z$ direction. The downstream boundary at $+z$ is adiabatic, and L_z is dynamically extended by adding segments to prevent any interaction of the shock front with the boundary.

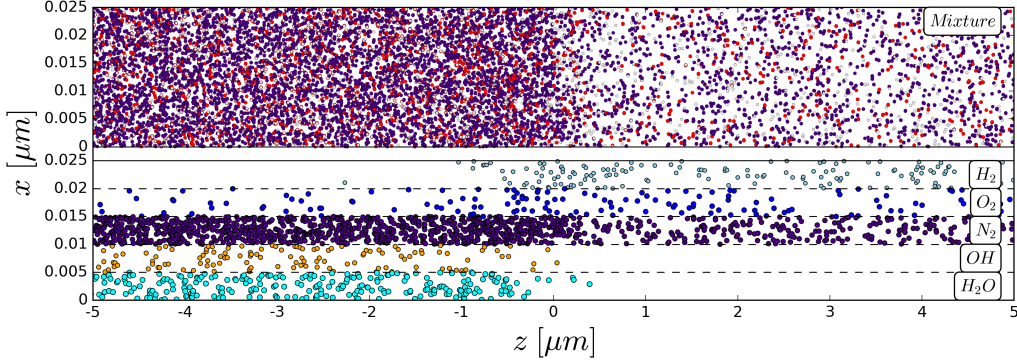


Fig. 1: Two-dimensional slice across the shock front region of the molecular distribution in the $\text{H}_2/\text{O}_2/\text{N}_2$ mixture for $\phi = 0.7$. In the upper panel, all species are displayed in a combined view with each atom plotted individually. Hydrogen atoms are shown in light gray, oxygen in red, and nitrogen in indigo. In the lower panel, the distributions of H_2 , O_2 , N_2 , OH , and H_2O are shown, with each molecule represented by a single point located at its center of mass.

We compare H_2/O_2 case at equivalence ratio $\phi = 1.0$ from our previous study [1] and two new ternary $\text{H}_2/\text{O}_2/\text{N}_2$ cases at $\phi = 1.0$ and $\phi = 0.7$. For the ternary mixtures and to mimic air-like conditions, the dilution is fixed by a four-to-one nitrogen-oxygen ratio, i.e., $Z_{\text{N}_2}^{\text{up}} = 4Z_{\text{O}_2}^{\text{up}}$ in the initial, unshocked state. The initial mole fractions then satisfy $Z_{\text{H}_2}^{\text{up}}/Z_{\text{O}_2}^{\text{up}} = 2\phi$, $Z_{\text{H}_2}^{\text{up}} + Z_{\text{O}_2}^{\text{up}} + Z_{\text{N}_2}^{\text{up}} = 1$, from which, for the binary case ($Z_{\text{N}_2}^{\text{up}} = 0$), $Z_{\text{H}_2}^{\text{up}} = 2\phi/(1 + 2\phi)$, $Z_{\text{O}_2}^{\text{up}} = 1/(1 + 2\phi)$, and, for the ternary case ($Z_{\text{N}_2}^{\text{up}} = 4Z_{\text{O}_2}^{\text{up}}$), $Z_{\text{O}_2}^{\text{up}} = 1/(2\phi + 5)$, $Z_{\text{H}_2}^{\text{up}} = 2\phi/(2\phi + 5)$, and $Z_{\text{N}_2}^{\text{up}} = 4/(2\phi + 5)$.

The piston moves at a constant speed v_p . For each composition, v_p is chosen such that, for the initial mixture treated as an ideal and nonreactive gas, the normal-shock relations yield a nominal $M = 4$. The effective Mach number of the reactive gases M_{eff} subsequently differs from 4 and is measured a posteriori in the shock frame. The corresponding value of M_{eff} is reported in Table 1.

Table 1: Set of cases and nominal parameters.

Mixture	ϕ	$Z_{\text{N}_2}^{\text{up}}/Z_{\text{O}_2}^{\text{up}}$	M_{eff}
H_2/O_2	1.0	0	4.94
$\text{H}_2/\text{O}_2/\text{N}_2$	1.0	4	4.65
$\text{H}_2/\text{O}_2/\text{N}_2$	0.7	4	4.61

Field quantities are averaged by one-dimensional binning along z in slices of fixed thickness. Tracking the position of the maximum mixture velocity in time yields the shock speed v_s . Number densities by species and derived quantities are obtained by molecular counting within each slice. Species identification relies on the analysis of *ReaxFF* bond orders.

3. Results

Figure 1 shows a two-dimensional slice of the atomic distribution centered on the shock front for the $\text{H}_2/\text{O}_2/\text{N}_2$ case with $\phi = 0.7$. The upstream mixture is located on the right, the shocked gas on the

left, and a transition layer identifies the front through a local increase in density. Moreover, the lower panels show that the reactants (H_2 , O_2) are progressively consumed in the post-shock region, while radical and product species (OH , H_2O) appear and grow in concentration, whereas nitrogen remains present across the entire domain as a weakly reactive diluent. Building on this observation, Subsection 3.1 examines the shock front structure and its dependence on composition, and Subsection 3.2 quantifies reaction progress in the shock front region.

3.1. Shock structure

Figure 2 shows normalized distributions of the major species along the z -axis for the three different mixtures, with trace species omitted for clarity. Although only minor amounts of NO_x are detected here (on the order of $\mathcal{O}(10)$ ppm), it is well established that combustion under sufficiently high temperature and residence time conditions can generate significant NO_x levels. The upstream mixture at rest is on the right, and the shock propagates from left to right, compressing and heating the gas. Reactant species exhibit a sharp elevation at the shock front, followed downstream by a decline as products emerge. Across all cases, the separation of the species shock fronts is observed. Although H_2O and OH emerge in close proximity near the O_2 shock front, the OH profile exhibits a delay before rising in the N_2 -diluted cases, with the delay more pronounced at $\phi = 1$. A possible explanation is the stronger conversion of newly formed OH into H_2O in the N_2 -diluted cases, which would limit the early accumulation of OH .

Figure 3 shows streamwise velocity profiles of reactants normalized by the piston speed. In the N_2 -diluted cases, the large oscillations of the H_2 arise due to the low sampling in regions where H_2 is nearly exhausted, as it can be observed in the species distributions in Figure 2. As expected the order of species follows molecular mass [14–16], i.e., H_2 leads, followed by N_2 , then O_2 . The magnitude of the separation grows with the mass disparity. Consequently, the

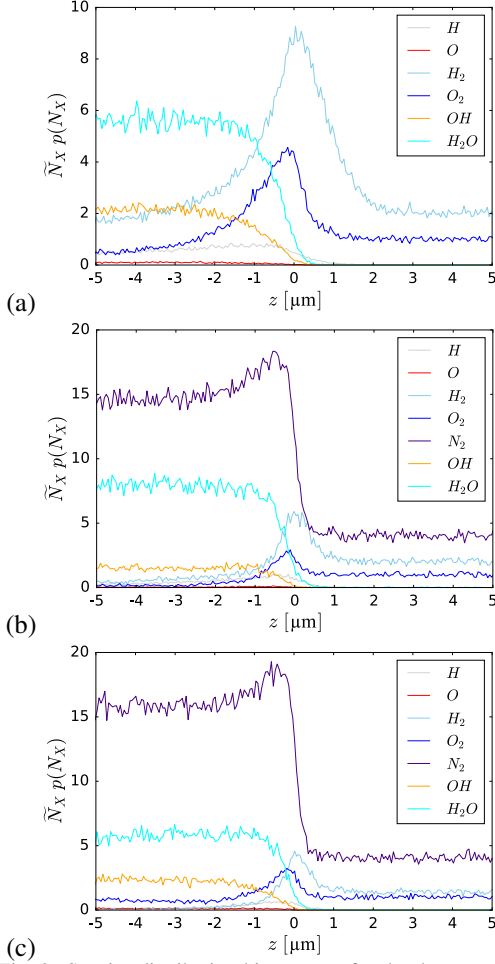


Fig. 2: Species distribution histograms, for the three cases in the same order as Table 1, where $p(N_X)$ represents the probability density of species X at position z , weighted by the total atoms per species $\tilde{N}_X = N_{X,m} N_{X,a}$, with $N_{X,m}$ being the total number of molecules of X and $N_{X,a}$ the number of all atoms per the molecule. To enable cross-case comparison, all profiles are normalized such that the upstream O_2 weighted density equals unity. Only the major species are shown, i.e., species present in negligible amounts are omitted.

N_2 and O_2 fronts remain close because their molecular masses are similar (28 and 32 g/mol, respectively), whereas the much lighter H_2 (2 g/mol) leads farther upstream. The separation between the H_2 shock front and the heavier-species fronts is larger in the undiluted binary case than in the two ternary cases. We attribute this primarily to the lower H_2 mole fraction together with the larger mass of the bath molecules in the diluted mixtures. A similar reduction of the front offset would be expected in a fuel-lean binary H_2/O_2 mixture. At the front, the combined effect of the reduced H_2 concentration and heavier bath molecules shortens the collisional stopping length of

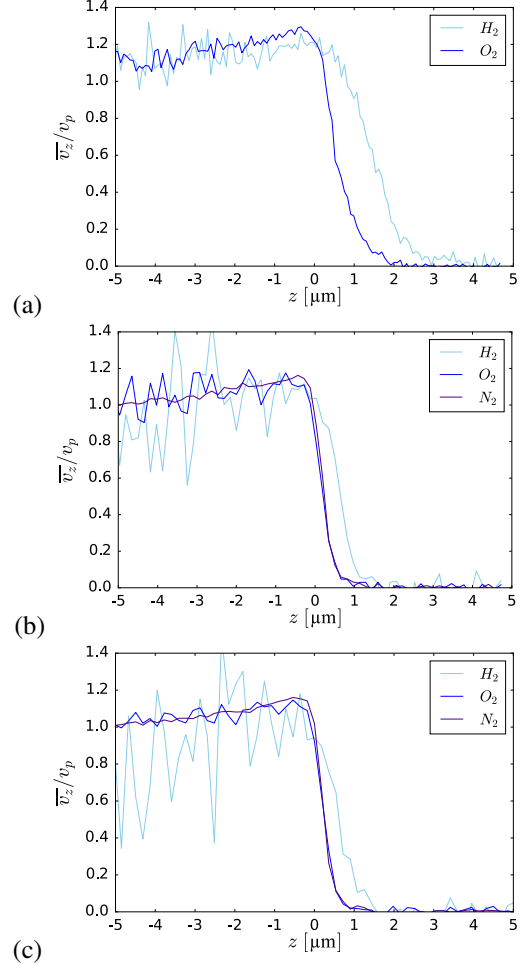


Fig. 3: Streamwise velocity profiles of reactants normalized by the piston speed in the same order as Table 1. Profiles are obtained by one-dimensional binning along the z -axis.

an H_2 molecule that diffuses upstream, as collisions with O_2 or N_2 remove a larger fraction of its momentum than H - H encounters and occur more frequently. Equivalently, the effective diffusivity of H_2 is smaller when its mole fraction is low and the mean bath molecule is heavy, which limits upstream penetration and reduces the front separation. Consequently, the N_2 and O_2 fronts remain close, and the H_2 front is less displaced upstream than in the undiluted binary case. Thus, the sharper velocity rise may enhance collisions in the ternary cases because particles do not accelerate gradually, thereby promoting collision-induced dissociation of the reactants. Note that these trends follow from purely kinematic and transport effects, i.e., mass disparity, number density, and momentum exchange, rather than chemistry. An inert gas mixture with the same composition and Mach number would exhibit the same ordering and reduced offsets of the species fronts.

3.2. Reaction progress in the shock front region

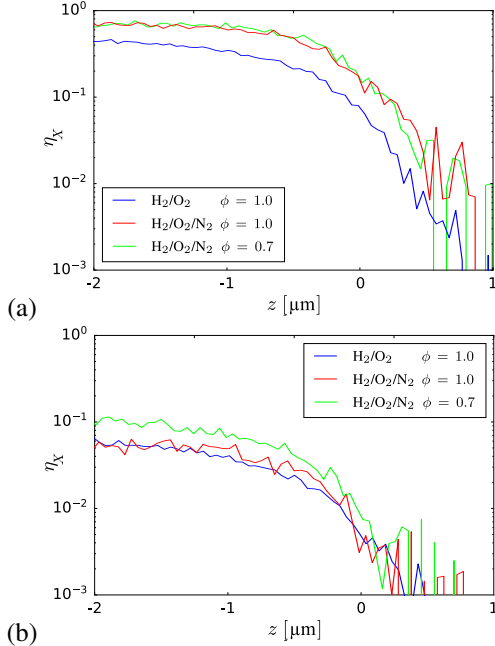


Fig. 4: Streamwise profiles of the reaction progress variable in the shock front region for (a) $\eta_{\text{H}_2\text{O}}$ and (b) η_{OH} for the three mixtures, shown in the same spatial reference frame for both subfigures.

For each species X , we define the reaction progress variable at a given location as

$$\eta_X = \frac{Z_X}{Z_{\text{H}_2}^{\text{up}}} \frac{n_H(X)}{2}, \quad (1)$$

where Z_X is the mole fraction of species X at the given location, $Z_{\text{H}_2}^{\text{up}}$ is the upstream H_2 mole fraction, and $n_H(X)$ is the number of hydrogen atoms per molecule X . Note that $Z_{\text{O}_2}^{\text{up}}$ is omitted because for $\phi \leq 1$, H_2 is the limiting species.

Figure 4 shows streamwise profiles of the reaction progress variables for OH and H_2O in the shock front region for the three mixtures. The reaction progress of H_2O increases in the presence of N_2 relative to the binary mixture. In contrast, η_{OH} is comparable to the binary value in the ternary stoichiometric case and higher in the lean case, consistent with the fact that under lean conditions limited hydrogen availability leaves a non-negligible radical pool. Considering H_2O and OH jointly, the lean ternary case exhibits a greater overall advancement of hydrogen conversion at the front, consistent with the expectation that a lower H_2 mole fraction promotes more frequent dissociation and recombination events, while remaining bounded by the equivalence ratio constraint.

As noted in the previous section, the magnified view centered on the shock region reveals a delay between the first appearance of OH and its subsequent

increase. This delay is larger in the ternary mixtures than in the binary case, suggesting early conversion of OH into H_2O as it forms at the shock front.

In our previous H_2/O_2 study of subcontinuum shock structure, dissociation was localized near the O_2 shock front [1]. The present results imply that the addition of N_2 modifies collisional characteristics in the shock front region. Owing to its larger molecular mass and strong bond energy, N_2 acts as an third body nearly nonreactive (under these conditions) collider at the shock front. As a consequence, N_2 can enhance collision-induced dissociation of H_2 and O_2 and promote nonreactive energy transfer without consuming radicals itself. In addition, the sharper velocity rise at the front produces stronger collisions, thereby increasing molecular dissociation. The net effect is an increase of $\eta_{\text{H}_2\text{O}}$ in the presence of nitrogen despite the thermal-dilution penalty.

4. Conclusion

We have shown, using reactive MD, that mixture composition is a key factor on subcontinuum reactive shock structure in gaseous $\text{H}_2/\text{O}_2/\text{N}_2$. All mixtures exhibit species-separated shock fronts, ordered by molecular mass, with H_2 leading, followed by N_2 and O_2 . The front offset is largest in the undiluted binary mixture and decreases with nitrogen dilution, which we attribute primarily to the lower upstream H_2 mole fraction and the larger mass of the surrounding bath molecules. At the front, these factors shorten the collisional stopping length and reduce the effective upstream penetration of H_2 , a purely kinematic and transport mechanism independent of chemical reactions. Near the O_2 front, OH and H_2O emerge in close proximity. In the diluted mixtures, OH exhibits a finite delay before increasing, plausibly due to early conversion of OH into H_2O at the front.

To quantify local chemical advancement, we evaluate reaction progress across the shock front. The conversion toward H_2O increases in the presence of nitrogen, exceeding the binary case. In contrast, the OH progress is comparable to the binary value at $\phi = 1$ and larger at $\phi = 0.7$ due to the hydrogen-lean conditions which does not allow a complete reaction of radicals. In addition, the sharper velocity rise in the diluted cases intensifies collisions at the front and, together with the nearly nonreactive third-body role of N_2 , promotes collision-induced dissociation and accelerates local progress toward H_2O . Under hydrogen-lean conditions where H_2 becomes limiting, this appears as sustained OH production. Overall, these results clarify how dilution and equivalence ratio set the interplay between shock structure and early reaction at subcontinuum scales, and they provide molecule-resolved benchmarks for mesoscale and continuum modeling of hydrogen-air detonations.

Declaration of competing interest

The authors declare that they have no known com-

peting financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] T. Maurel-Oujia, K. Maeda, Subcontinuum structures of reactive shock waves in gaseous H_2/O_2 mixtures, *Physical Review Fluids* 10 (10) (2025) 103201.
- [2] H. D. Perkins, D. E. Paxson, Summary of pressure gain combustion research at nasa, Tech. rep. (2018).
- [3] P. A. Strakey, D. H. Ferguson, Experimental measurements and cfd predictions of nox emissions from a water-cooled rotating detonation engine, in: *Turbo Expo: Power for Land, Sea, and Air*, Vol. 86953, American Society of Mechanical Engineers, 2023, p. V03AT04A005.
- [4] M. Oevermann, Numerical investigation of turbulent hydrogen combustion in a scramjet using flamelet modeling, *Aerospace science and technology* 4 (7) (2000) 463–480.
- [5] T. Yamada, A. K. Hayashi, E. Yamada, N. Tsuboi, V. E. Tangirala, T. Fujiwara, Detonation limit thresholds in H_2/O_2 rotating detonation engine, *Combustion Science and Technology* 182 (11-12) (2010) 1901–1914.
- [6] S. Frolov, V. Aksenov, V. Ivanov, I. Shamshin, Large-scale hydrogen–air continuous detonation combustor, *International Journal of Hydrogen Energy* 40 (3) (2015) 1616–1623.
- [7] A. K. Hayashi, N. Tsuboi, E. Dzieminska, Numerical study on jp-10/air detonation and rotating detonation engine, *AIAA journal* 58 (12) (2020) 5078–5094.
- [8] R. Grosseuvres, A. Comandini, A. Bentaib, N. Chaumeix, Combustion properties of $\text{H}_2/\text{N}_2/\text{O}_2/\text{steam}$ mixtures, *Proceedings of the Combustion Institute* 37 (2) (2019) 1537–1546.
- [9] H. M. Mott-Smith, The solution of the boltzmann equation for a shock wave, *Physical Review* 82 (6) (1951) 885.
- [10] A. P. Thompson, H. M. Aktulga, R. Berger, D. S. Bolintineanu, W. M. Brown, P. S. Crozier, P. J. In't Veld, A. Kohlmeyer, S. G. Moore, T. D. Nguyen, et al., LAMMPS—a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales, *Computer physics communications* 271 (2022) 108171.
- [11] A. C. Van Duin, S. Dasgupta, F. Lorant, W. A. Goddard, Reaxff: a reactive force field for hydrocarbons, *The Journal of Physical Chemistry A* 105 (41) (2001) 9396–9409.
- [12] K. Chenoweth, A. C. Van Duin, W. A. Goddard, Reaxff reactive force field for molecular dynamics simulations of hydrocarbon oxidation, *The Journal of Physical Chemistry A* 112 (5) (2008) 1040–1053.
- [13] C. Ashraf, A. C. Van Duin, Extension of the reaxff combustion force field toward syngas combustion and initial oxidation kinetics, *The Journal of Physical Chemistry A* 121 (5) (2017) 1051–1068.
- [14] F. S. Sherman, Shock-wave structure in binary mixtures of chemically inert perfect gases, *Journal of Fluid Mechanics* 8 (3) (1960) 465–480.
- [15] G. Bird, The structure of normal shock waves in a binary gas mixture, *Journal of Fluid Mechanics* 31 (4) (1968) 657–668.
- [16] D. Madjarević, M. Pavić-Čolić, S. Simić, Shock structure and relaxation in the multi-component mixture of euler fluids, *Symmetry* 13 (6) (2021) 955.